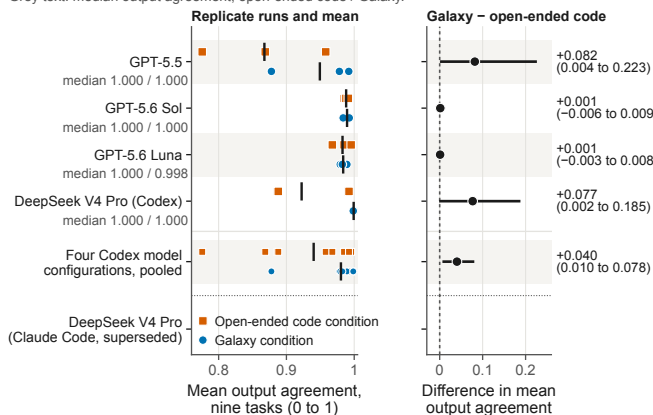
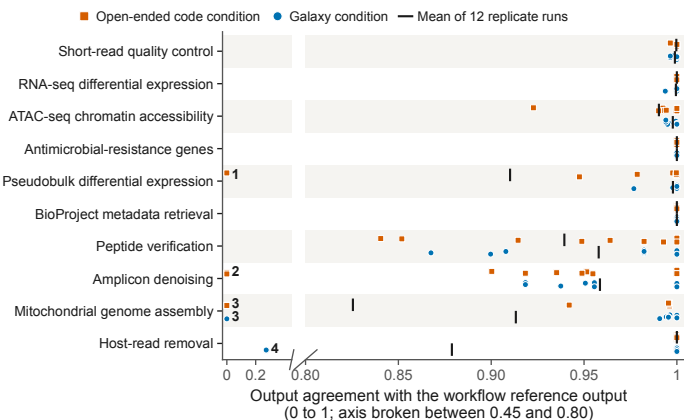


a IWC: mean output agreement was higher in the Galaxy condition on the workflow-derived benchmark

Nine tasks scored in both conditions. Pooled mean: open-ended code 0.940, Galaxy 0.980.
Grey text: median output agreement, open-ended code / Galaxy.



b IWC: six of ten tasks reached near-perfect Galaxy-condition mean output agreement; every run below 0.5 has a traced cause

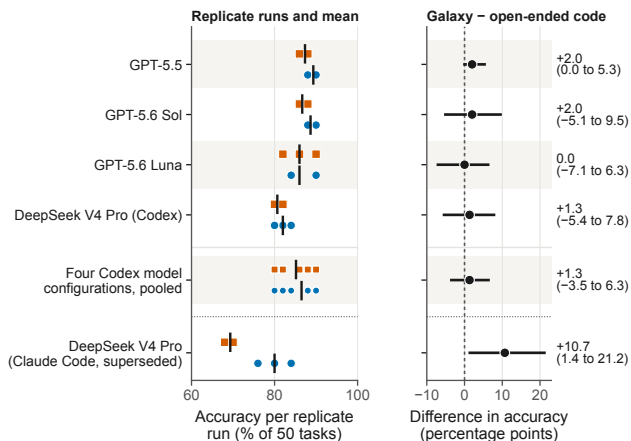


Runs with output agreement below 0.5 (numbers on the plot)

- Open-ended code only: hand-written multiple-testing correction was wrong (DeepSeek V4 Pro (Codex), replicate run 3); installed Galaxy tools return library-computed adjusted P values
- Open-ended code only: sample names lost capital letters when read from directory names (GPT-5.5, replicate runs 1 and 3); Galaxy collections keep them
- Wrong contig submitted: GPT-5.5 replicate run 1 in both conditions; DeepSeek V4 Pro (Codex) replicate run 2, open-ended code condition
- Galaxy condition, GPT-5.6 Luna, replicate runs 1 and 3: score conflict; correct BWA-MEM output scored against the Bowtie2 reference

c BixBench-Verified-50: similar accuracy in both execution conditions

Pooled accuracy, four Codex model configurations: open-ended code 85.2%, Galaxy 86.5%.



d CompBioBench: similar reported benchmark scores in both execution conditions

Mean of replicate runs, four Codex model configurations: open-ended code 86.7, Galaxy 86.9 of 100. No interval: the archive keeps no per-run grades.

